DD-Mamba: Dual-Domain State-Space Forecasting with a Linear Time Backbone and a Conditional Spectral Pathway

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Abstract. From power grids to road networks, long-term multivariate forecasting must capture local time-domain dynamics, periodic structure that is compact in the frequency domain, and cross-channel dependencies whose strength varies enormously across datasets: the mean absolute inter-channel correlation spans 0.31 on ETTh1 to 0.91 on Solar. Yet most deep forecasters hard-wire one choice on every axis — a single representation, a single channel regime, uniform capacity for all data. We present DD-Mamba, a dual-domain state-space framework that makes these choices testable: a decomposition-based time branch and a learned spectral branch each produce a forecast, fused by a convex gate; a DLinear-style backbone with zero-initialized corrections makes the initial prediction linear in the instance-normalized window; and a bidirectional Mamba over variates is a per-dataset switch guided by measured channel coupling. Across nine look-back-96 benchmarks (three seeds), DD-Mamba beats reported baselines beyond a 0.005 descriptive tolerance on Solar, ETTh1, and ETTh2 — the first two also clear a wider 0.01 cross-paper caution — ties the leading reported results on four datasets, and trails on Traffic and Exchange. Paired ablations track the correlation split: the variate mixer helps Weather yet hurts weakly coupled ETTh1, and RevIN helps Weather yet hurts Solar. A 27-cell branch-ablation matrix finds the frequency path significant on four datasets at the 96-step horizon, fading at longer ones, and the branches redundant elsewhere. The evidence maps which data characteristics reward which components: conditional placement, not universal gain.

Keywords: Time series forecasting · State-space models · Mamba · Frequency domain · Multivariate analysis

1 Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [27, 22, 28], patch-based attention [17], inverted variate-token attention [13], and, most recently, selective state-space models (SSMs) built on Mamba [8, 20], which offer linear-time sequence modeling with input-dependent dynamics.

Three empirical gaps complicate this progress. **Gap 1: deep models discard strong linear maps.** A per-component linear projection from the look-back window to the horizon (DLinear [25]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [23]) can be competitive with much larger models on standard benchmarks. Without such a direct pathway, a deep architecture must learn any linear behavior implicitly, which may be inefficient on small datasets. **Gap 2: the useful inductive bias differs by domain.** Local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain, yet models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [24] — do not expose both views as separately testable forecast paths. **Gap 3: cross-variate capacity can help or hurt.** Cross-variate mixing is effective on several high-dimensional benchmarks [13, 20, 26], but the underlying data property it exploits — how strongly channels are actually correlated — differs by a factor of three across the standard benchmarks (Table 1), and our controlled results show the same module can overfit a weakly coupled, low-channel dataset. This motivates treating the mixer as an explicit, data-diagnosable option rather than a universal setting.

We answer these three gaps with DD-Mamba (Fig. 1), mapping each to one structural remedy: *(i)* a DLinear-style time backbone with zero-initialized non-linear corrections; *(ii)* parallel time and frequency forecasts, including an optional dormant FITS-style spectral map, fused by a convex gate; and *(iii)* a bidirectional variate mixer switched on or off per dataset. Paired ablations test these choices below; only effects whose confidence intervals exclude zero are treated as supported.

Concretely, the time branch combines a DLinear-style pathway [25] with a zero-initialized nonlinear correction, while the frequency branch combines learned filtering with an optional zero-initialized FITS-style spectral pathway [23]. The resulting model starts as a linear map of the instance-normalized window (the RLinear-class structure [11, 10]); a controlled test finds zero initialization accuracy-neutral. The branch forecasts form a learned convex combination. The strongest ablation evidence concerns *placement*: removing the variate mixer hurts Weather whereas adding it hurts ETTh1, each with a paired CI excluding zero, and normalization shows the same two-sided pattern. A 27-cell branch-ablation matrix then locates where each domain matters: the frequency path is individually significant on four datasets at $H=96$ (Solar, ETTm1, Electricity, Exchange) but fades at longer horizons, the time branch is necessary on ETTm1 at every horizon, and the branches are redundant on ETTh1, ETTh2, Weather, and Traffic — so the dual-domain claim is supported, but conditional on both the dataset and the horizon.

Contributions. (1) Two-sided, cross-dataset evidence that state-space capacity and normalization choices can help or hurt — the case for per-dataset placement over universal defaults; **(2)** a DLinear-style time backbone with zero-initialized deep corrections and an optional dormant spectral pathway, with a controlled initialization ablation; **(3)** a dual-domain architecture whose output is a gated convex combination of independently testable branch forecasts, supported by a 27-cell branch-ablation matrix that maps where — by dataset and horizon — each branch is necessary or redundant; and **(4)** three-seed results on nine benchmarks plus paired ablations, including statistically supported negative and dataset-dependent findings.

2 Related Work

Time series analysis problems. Deep models now serve the full spectrum of time series mining tasks — forecasting, classification, anomaly detection, and imputation — often through shared general-purpose backbones such as TimesNet [21], whose representations transfer across tasks. Long-term multivariate *forecasting*, our setting, is the most benchmark-standardized of these problems: fixed look-back, multi-horizon evaluation, and shared splits [27, 13] make it the natural arena for controlled architectural evidence, and findings about which structures a dataset rewards typically carry over to the sibling tasks that reuse forecasting backbones.

Temporal-dependency models. One line models dependencies along the time axis only. Transformer variants adapt attention to long sequences via sparsity, decomposition, and frequency enhancement [27, 22, 28], patching [17], or explicit de-/re-stationarization [14]. A second family shows that *linear* maps remain competitive: DLinear/NLinear [25], RLinear with reversible instance normalization [11, 10], the MLP-based TiDE [6] and TimeMixer [19], and the residual-stacked N-BEATS/N-HiTS [18, 3], our closest residual-forecasting precedents. A third works in the spectrum: FreTS [24] learns frequency-domain MLPs and FITS [23] forecasts with one complex-linear interpolation of the low-passed spectrum. Decomposition-based designs (STL [4], SCINet [12], and the seasonal-trend-dispersion view of STD [7]) make component structure explicit; RevIN, used here as a per-dataset switch, normalizes windows reversibly but neither extracts nor forecasts dispersion, and we claim no STD-style decomposition.

Cross-dimensional (spatial–temporal) models. A second line treats the variate dimension as a first-class axis. Crossformer [26] is the classic Transformer of this type: it embeds the series into a two-dimensional patch grid and attends over *both* the temporal and the cross-variate (spatial) dimension. iTransformer [13] inverts the axes and attends over variate tokens only; SOFTS [9] reaches linear-complexity channel interaction through a centralized core, and ModernTCN [15] interleaves temporal and cross-variable convolutions. Selective state-space models bring linear-time scans to the same axes: Mamba [8, 5] over time, and S-Mamba [20], TimeMachine [1], and FMamba [16] over variate tokens. DD-Mamba sits in this lineage — causal Mamba over time, bidirectional Mamba over variates and frequency bins — but treats the spatial dimension as a *switchable* module rather than a fixed design commitment.

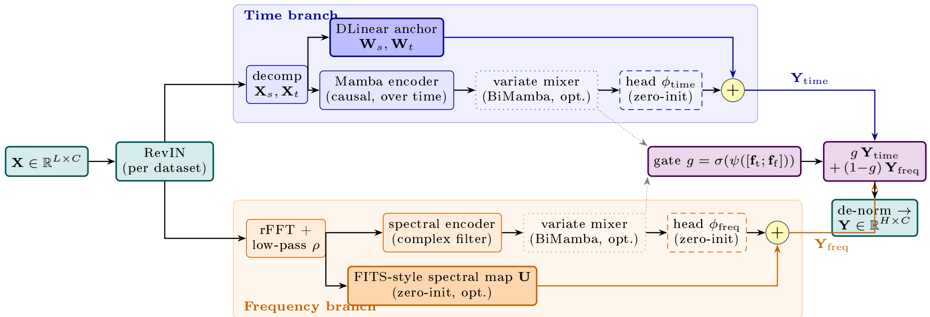

Fig. 1. DD-Mamba overview, color-coded by role: the **normalization** wrapper (teal), the **time branch** (blue), the **frequency branch** (orange), and the **convex-gate fusion** (violet). In each time branch, a solid-filled DLinear-style backbone runs in parallel with a zero-initialized nonlinear head (dashed). The optional frequency-domain map and its nonlinear head are also zero-initialized, so the model begins as a scaled linear map of the instance-normalized window. Dotted boxes are per-dataset switches; dotted grey arrows carry branch features to the gate, whose convex weight g mixes the two coloured forecast paths $\mathbf{Y}_{\text{time}}$ and $\mathbf{Y}_{\text{freq}}$.

Research gap. Across both lines, three commitments are usually hard-wired. (i) Each model adopts one representation — time steps *or* spectra — so the other view's contribution is never separately testable. (ii) Cross-dimensional models such as Crossformer apply the same spatial–temporal capacity to every dataset, although channel coupling varies enormously across benchmarks (mean absolute cross-channel Pearson correlation ranges from 0.31 on ETTh1 to 0.91 on Solar; Table 1). (iii) Deep designs typically discard the linear maps that remain near state-of-the-art, relearning them implicitly. No prior work, to our knowledge, pairs a per-component switchboard with controlled, seed-paired evidence of *where* — for which datasets, which correlation regimes, and which horizons — each structure pays. DD-Mamba is built to answer exactly that question: independently testable time/frequency forecast paths, a switchable variate mixer, zero-initialized corrections in the spirit of ReZero [2] (retained here as an interpretable initialization convention, shown accuracy-neutral), and a 27-cell branch-ablation matrix connecting data characteristics to component value.

3 Method

3.1 Problem Formulation and Normalization

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L steps and C variates, the task is to predict the next H steps $\mathbf{Y} \in \mathbb{R}^{H \times C}$. Following RevIN [10], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, and de-normalize the final forecast. Instance normalization is treated as a per-dataset switch rather than a universal default; Sect. 4.3 shows why.

3.2 Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [22, 25] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s\mathbf{X}_s + \mathbf{W}_t\mathbf{X}_t \in \mathbb{R}^{H\times C}, \qquad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H\times L}, \tag{1}$$

i.e., a DLinear-style map [25] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [8] (Fig. 2a). Following the zero-order-hold discretization of [8], the continuous parameters $(\boldsymbol{\Delta}, \mathbf{A}, \mathbf{B})$ become $\bar{\mathbf{A}}_k = \exp(\boldsymbol{\Delta}_k\mathbf{A})$ and $\bar{\mathbf{B}}_k = \boldsymbol{\Delta}_k\mathbf{B}_k$, and the layer runs the recurrence

$$\mathbf{h}_k = \bar{\mathbf{A}}_k\,\mathbf{h}_{k-1} + \bar{\mathbf{B}}_k\,u_k, \qquad y_k = \mathbf{C}_k^\top\mathbf{h}_k + D\,u_k, \tag{2}$$

scanned causally over the L steps and applied independently to each of the d_{in} feature dimensions: per dimension, the state is $\mathbf{h}_k \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^N$ is diagonal, the input u_k is scalar, and $(\boldsymbol{\Delta}_k \in \mathbb{R},\ \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$ are produced from u by learned linear maps — the input-dependent selectivity of [8]. The last state, summed with a whole-window linear embedding $\mathbf{W}_e\tilde{\mathbf{x}}^{(c)}$ (one per variate, as in [20]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C\times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \qquad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \tag{3}$$

Because ϕ_{time} is zero-initialized, the branch's output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express.

3.3 Frequency Branch: Spectral Filtering and a Dormant Spectral Pathway

The frequency branch applies the real FFT along time, $\mathbf{Z} = \text{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C\times F}$, $F = \lfloor L/2\rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i \in \mathbb{C}^{F\times F}$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top$ along the frequency axis (a bidirectional Mamba over the bin sequence is an input-dependent alternative). The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C\times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS-style spectral pathway (optional, per dataset). A complex linear map $\mathbf{U} \in \mathbb{C}^{F'\times F}$ with $F' = \lfloor (L{+}H)/2\rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C\times F'}$), maps the (low-passed) window spectrum to that of a length-$(L{+}H)$ series, whose inverse transform's last H samples are the branch's linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\text{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[\,L{:}\,]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \qquad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \tag{4}$$

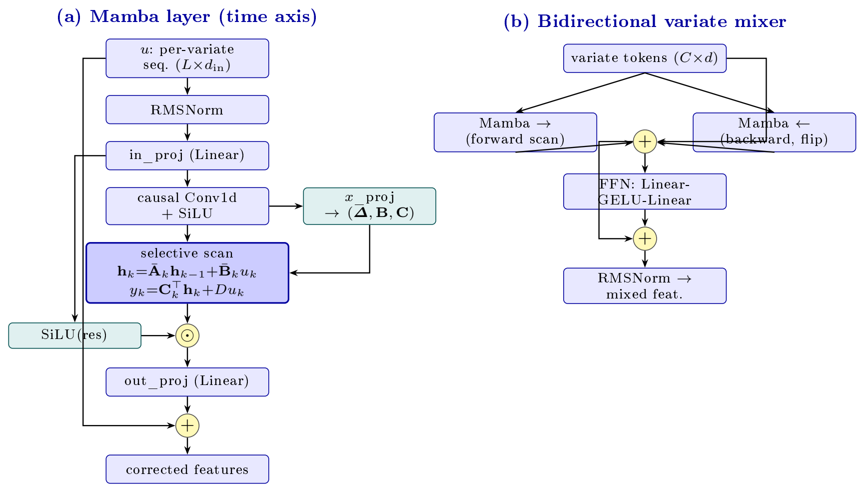

Fig. 2. Internal structure of DD-Mamba's two selective-state-space blocks (cf. the blue boxes in Fig. 1). **(a)** The causal Mamba layer used as the time-axis encoder (Eq. (2)): an input-dependent selective scan, gated by a SiLU branch and wrapped in a residual, with $(\boldsymbol{\Delta}, \mathbf{B}, \mathbf{C})$ predicted per step from the input. **(b)** The bidirectional mixer used over the *variate* dimension (Sect. 3.4): forward and backward scans of the C channel tokens summed in one residual stream, then a position-wise feed-forward block (the S-Mamba design [20]).

This pathway uses the frequency interpolation idea of FITS [23], but is not a fitted FITS model: $\mathbf{U}$ is *zero-initialized*, so the frequency branch is silent at initialization (the length-L and length-$(L+H)$ frequency grids do not align, so no canonical identity init exists, and a random $\mathbf{U}$ would inject noise during the highest-learning-rate epochs). With the time head, spectral map, and gate all zero or constant at init, the end-to-end map in the normalized space is the linear $g_0 \cdot \mathbf{Y}_{\text{lin}}$; composed with RevIN this is the RLinear-class structure [11, 10] — linear in the normalized window, by construction not fitted. The spectral map is enabled per dataset (Table 2); where disabled, the frequency branch remains active through its encoder and forecast head.

3.4 Cross-Channel Mixing over Variates

Channel independence is a strong baseline [17, 25] but leaves accuracy on the table when variates are coupled [13, 20, 26]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the C per-channel features, followed by a position-wise feed-forward block (Fig. 2b). Bidirectionality is essential: channel order carries no causal direction, so each layer sums a forward and a reversed scan in one residual stream. Whether such mixing *should* pay is a property of the data: the mean absolute pairwise Pearson correlation (PCC) between channels on the training split spans 0.31–0.35 on the weakly coupled ETT family but 0.50–0.91 on Electricity, Traffic, and Solar (Table 1). We therefore expose the mixer as a per-dataset switch — on for strongly coupled, many-channel benchmarks, off for weakly coupled, few-channel ones (per-dataset settings in Sect. 4.1) — and test the decision in both directions in Sect. 4.3.

3.5 Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1-g) \odot \mathbf{Y}_{\text{freq}}, \qquad g = \sigma\big(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])\big) \in (0,1)^{C\times 1}, \quad (5)$$

with ψ applied row-wise (per variate), followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value reports each branch's convex mixing weight — a lightweight proxy for its contribution, not a causal attribution. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little — by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with its bias set so that training starts gate-open toward the time branch ($g_0 \approx 0.9$; exact value in Sect. 4.1): the time branch is linear at initialization while the frequency branch is silent, and an even initial mix would halve the initial linear forecast. This is an initialization convention; empirical gate distributions are not available in the present study.

3.6 Training

We minimize MSE with AdamW under a halving schedule ($\eta_e = \eta_0 \cdot 2^{-(e-1)}$). All runs use 10 epochs with best-checkpoint early stopping. This schedule was held fixed across our runs but was not compared with cosine, warm-up, or longer-budget alternatives; schedule sensitivity therefore remains a potential confound.

4 Experiments

4.1 Setup

Datasets and protocol. We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [13, 20]: look-back 96, horizons 96/192/336/720. ETT uses the canonical 12/4/4-month border split, the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE.

Baselines. We compare against the strongest published results under this protocol: S-Mamba [20], iTransformer [13], PatchTST [17], DLinear [25], and TimesNet [21]. Numbers are quoted from [20] rather than reproduced: the protocol is identical, but training pipelines, seeds, and implementations differ across works, so gaps below the descriptive tolerance in Sect. 4.1 relative to quoted numbers should be read with caution (flagged in the text); a unified rerun is planned for the camera-ready appendix.

Table 1. Benchmark datasets, their cross-channel coupling, and where DD-Mamba places its capacity. $\overline{|r|}$: mean absolute pairwise Pearson correlation between channels on the training split (— = raw file unavailable in our anonymized build environment).

Dataset	Variates	Steps	Granularity	$\overline{\lvert r\rvert}$	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	0.31/0.35	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	0.31/0.35	off	0.24–0.36M
Weather	21	52,696	10 min	—	M=2, d=256	5.8M
Solar-Energy	137	52,560	10 min	0.91	M=1, d=128	0.96M
Electricity	321	26,304	1 hour	0.50	M=2, d=512	19.7M
Traffic	862	17,544	1 hour	0.58	M=2, d=512	19.7M
Exchange	8	7,588	1 day	0.48	off	0.09M

Implementation and Reproducibility. All experiments use PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090. Optimization is AdamW (grad clip 5.0), 10 epochs with the learning rate halved each epoch, early stopping on validation loss, and mixed precision with the selective-scan recurrence kept in full precision for numerical stability of the L-step state accumulation. The decomposition kernel is 25; Mamba blocks use expansion 2 and conv width 4 throughout; the fusion gate's bias is initialized to 2.2, giving $g_0 = \sigma(2.2) \approx 0.9$ (Sect. 3.5). Table 2 lists all per-dataset settings; configs and the ablation harness ship with the code. All main results are means over *three seeds* (2024–2026); the maximum per-entry std across the 36 dataset–horizon entries is 0.005 MSE, typically 0.001–0.003. The table records final empirical configurations, not a transferable capacity-selection rule: no common pre-registered search space or equalized tuning budget was used across all datasets. Consequently, capacity placement is reported as an empirical finding, and new datasets require validation-only selection of these switches.

Statistical methodology. We compare at two levels. *Within our framework* (ablations), every variant shares seeds with the reference, so we report *paired* per-seed differences with t-based 95% CIs (n=3) and treat a component as supported only when its CI excludes zero. These intervals are exploratory: n=3 limits power, and no family-wise correction is applied across ablations. *Against quoted baselines*, pairing is impossible, so 0.005 MSE is used only as a descriptive tolerance based on the largest standard deviation among our runs, not as a significance test. Our in-framework DLinear re-run shows why: through the identical pipeline on ETTh1 it averages 0.462/0.456 MSE/MAE versus its quoted 0.456/0.452 — a +0.006 implementation gap, the order of several margins in Table 3. Cross-paper comparisons therefore cannot support sub-0.01 distinctions; DD-Mamba's margin over the in-framework DLinear (0.439 vs. 0.462) is, by contrast, a same-pipeline comparison.

Table 2. Per-dataset hyperparameters. d: model width; time enc.: time-branch encoder (Mamba layers in parentheses); M: variate-mixer layers; N: SSM state size; ρ: spectral low-pass fraction (Sect. 3.3); spec. map: the *optional* FITS-style pathway of Eq. (4) — when "no", the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. †Solar adopts IN = no from the ablation finding (Sect. 4.3); its main results were re-measured under this configuration across all horizons and seeds.

Dataset	d	time enc.	M	N	ρ	spec. map	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5{\cdot}10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5{\cdot}10^{-4}$	32	$0.3/5{\cdot}10^{-4}$
ETTm1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTm2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2{\cdot}10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2{\cdot}10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no†	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no	yes	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5{\cdot}10^{-4}$

Table 3. Long-term forecasting results averaged over horizons $H \in \{96, 192, 336, 720\}$ with look-back L=96 (MSE/MAE; lower is better). DD-Mamba entries are means over three seeds (Table 4 reports per-horizon std). Baselines quoted from [20]. **Bold** marks the best result *and* every result within the descriptive tolerance (< 0.005, Sect. 4.1) of it; this formatting is not a cross-paper significance claim; <u>underline</u> marks the next-best result.

Dataset	DD-Mamba (ours)	S-Mamba	iTransformer	PatchTST	DLinear	TimesNet
ETTh1	**0.439**/**0.434**	0.455/0.450	<u>0.454</u>/<u>0.447</u>	0.469/0.454	0.456/0.452	0.458/0.450
ETTh2	**0.374**/**0.401**	<u>0.381</u>/**0.405**	0.383/<u>0.407</u>	0.387/<u>0.407</u>	0.559/0.515	0.414/0.427
ETTm1	**0.386**/**0.398**	<u>0.398</u>/<u>0.405</u>	0.407/0.410	**0.387**/**0.400**	0.403/0.407	0.400/0.406
ETTm2	**0.280**/**0.325**	<u>0.288</u>/<u>0.332</u>	<u>0.288</u>/<u>0.332</u>	**0.281**/**0.326**	0.350/0.401	0.291/0.333
Weather	**0.248**/**0.276**	**0.251**/**0.276**	<u>0.258</u>/**0.278**	0.259/<u>0.281</u>	0.265/0.317	0.259/0.287
Electricity	**0.168**/**0.263**	**0.170**/**0.265**	<u>0.178</u>/<u>0.270</u>	0.205/0.290	0.212/0.300	0.192/0.295
Traffic	0.438/<u>0.281</u>	**0.414**/**0.276**	<u>0.428</u>/0.282	0.481/0.304	0.625/0.383	0.620/0.336
Exchange	0.363/**0.405**	0.367/<u>0.408</u>	<u>0.360</u>/**0.403**	0.367/**0.404**	**0.354**/0.414	0.416/0.443
Solar	**0.199**/**0.260**	0.240/<u>0.273</u>	<u>0.233</u>/**0.262**	0.270/0.307	0.330/0.401	0.301/0.319

4.2 Main Results

Against the quoted results and under the stated descriptive tolerance, DD-Mamba shows average-MSE gaps larger than 0.005 on Solar, ETTh1, and ETTh2 (Table 3; per-horizon values in Table 4), although only the Solar (0.034) and ETTh1 (0.015) margins also exceed the wider 0.01 cross-paper caution of Sect. 4.1; ETTh2's 0.007 lies between the two thresholds. The largest margin is on Solar (0.199 vs. iTransformer's 0.233; −17% vs. S-Mamba, with the normalization-free configuration of Sect. 4.3). On ETTm1 and ETTm2 it improves over the quoted S-Mamba result but ties PatchTST, and on Weather and Electricity the gap is below the tolerance. It trails DLinear on Exchange (by 0.009, above tolerance) and S-Mamba on Traffic (by 0.024), which we *hypothesize* reflects the weekly cycle exceeding the look-back. As all comparisons are to numbers *reported* in [20], we phrase the claim conservatively: DD-Mamba is competitive with re-

Table 4. DD-Mamba per-horizon results (mean ± std over three seeds), look-back L=96.

Dataset		H=96	H=192	H=336	H=720	Avg
ETTh1	MSE	0.380$\pm_{.001}$	0.431$\pm_{.002}$	0.474$\pm_{.004}$	0.470$\pm_{.003}$	0.439
	MAE	0.396$\pm_{.001}$	0.424$\pm_{.001}$	0.447$\pm_{.003}$	0.468$\pm_{.001}$	0.434
ETTh2	MSE	0.293$\pm_{.001}$	0.368$\pm_{.003}$	0.414$\pm_{.001}$	0.421$\pm_{.001}$	0.374
	MAE	0.344$\pm_{.002}$	0.391$\pm_{.000}$	0.426$\pm_{.000}$	0.441$\pm_{.001}$	0.401
ETTm1	MSE	0.320$\pm_{.001}$	0.366$\pm_{.002}$	0.398$\pm_{.001}$	0.462$\pm_{.001}$	0.386
	MAE	0.359$\pm_{.001}$	0.384$\pm_{.001}$	0.406$\pm_{.001}$	0.443$\pm_{.001}$	0.398
ETTm2	MSE	0.177$\pm_{.000}$	0.242$\pm_{.002}$	0.302$\pm_{.000}$	0.401$\pm_{.003}$	0.280
	MAE	0.260$\pm_{.001}$	0.303$\pm_{.001}$	0.340$\pm_{.000}$	0.398$\pm_{.002}$	0.325
Weather	MSE	0.162$\pm_{.000}$	0.212$\pm_{.000}$	0.269$\pm_{.001}$	0.352$\pm_{.003}$	0.248
	MAE	0.207$\pm_{.001}$	0.255$\pm_{.000}$	0.295$\pm_{.002}$	0.349$\pm_{.002}$	0.276
Electricity	MSE	0.140$\pm_{.000}$	0.157$\pm_{.001}$	0.174$\pm_{.001}$	0.200$\pm_{.004}$	0.168
	MAE	0.235$\pm_{.000}$	0.252$\pm_{.001}$	0.271$\pm_{.001}$	0.296$\pm_{.004}$	0.263
Traffic	MSE	0.402$\pm_{.002}$	0.424$\pm_{.001}$	0.441$\pm_{.001}$	0.485$\pm_{.003}$	0.438
	MAE	0.266$\pm_{.001}$	0.275$\pm_{.001}$	0.283$\pm_{.000}$	0.301$\pm_{.000}$	0.281
Exchange	MSE	0.086$\pm_{.000}$	0.179$\pm_{.001}$	0.330$\pm_{.002}$	0.857$\pm_{.001}$	0.363
	MAE	0.205$\pm_{.000}$	0.301$\pm_{.000}$	0.416$\pm_{.002}$	0.699$\pm_{.000}$	0.405
Solar	MSE	0.180$\pm_{.004}$	0.195$\pm_{.005}$	0.207$\pm_{.001}$	0.212$\pm_{.005}$	0.199
	MAE	0.237$\pm_{.003}$	0.259$\pm_{.003}$	0.270$\pm_{.001}$	0.274$\pm_{.005}$	0.260

sults reported under this protocol, its clearest margins on Solar and ETTh1, where the linear backbone and capacity restraint matter most and DD-Mamba uses $\approx$ 0.25M parameters — roughly an order of magnitude fewer than variate-attention models configured for these datasets.

4.3 Component Ablations

Our ablation harness flips exactly one switch of a dataset's full configuration, holding split, schedule, seeds, and every other hyperparameter fixed. We report its complete output on three datasets under their *final released configurations*, three seeds each at H=96: ETTh1 and Solar (Table 5) and Weather (Table 7), with the paired methodology of Sect. 4.1. Two consistency checks pass: the Solar "full" row reproduces the main-table Solar H=96 entry (0.180$\pm_{.004}$), and its w/o-instance-norm variant coincides with the full (RevIN-off) configuration, so that cell is omitted.

Primary finding: capacity choices can help or hurt. The most robust result in our studies is two-sided and spans datasets. Adding the variate mixer on ETTh1 *hurts* (+0.0077*) while removing it on Weather also hurts (+0.0104*); normalization shows the same two-sided pattern (removing it hurts Weather +0.0161*, while disabling it on Solar improved the re-measured main results from 0.228 to 0.199). Zero initialization is a *null* result — no significant difference on either dataset (−0.0012 ETTh1, −0.0046 Solar; CIs include zero) — so we make no accuracy claim for it, only the interpretable linear start (Sect. 3.2). The DLinear backbone and the learned gate versus averaging are within CIs everywhere; on Weather (Table 7) three components have unadjusted paired CIs excluding zero (normalization, the variate mixer, and the temporal Mamba, +0.0027, positive on all seeds). The recurring pattern — component value is dataset-dependent,

Table 5. Component ablations on ETTh1 and Solar-Energy (H=96, three seeds, final configurations): paired per-seed ΔMSE to the full model with t-based 95% CIs; * CI excludes zero. The spectral map is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

Variant	ETTh1: Δ [95% CI]	Solar: Δ [95% CI]
full (reference MSE)	0.3802±.001	0.1802±.004
time branch only (freq removed)	+0.0034 [−0.009, +0.016]	+0.0114 [+0.001, +0.021]*
freq branch only (time removed)	−0.0007 [−0.004, +0.003]	+0.0051 [−0.013, +0.023]
sum fusion	−0.0026 [−0.007, +0.002]	+0.0066 [−0.005, +0.018]
w/o instance norm	+0.0061 [−0.001, +0.013]	—
w/o linear backbone	−0.0031 [−0.009, +0.003]	+0.0081 [−0.019, +0.035]
random init (vs. zero-init)	−0.0012 [−0.005, +0.002]	−0.0046 [−0.022, +0.013]
spectral map removed / added	+0.0042 [−0.018, +0.026]	+0.0088 [−0.002, +0.019]
variate mixer added / removed	+0.0077 [+0.004, +0.012]*	+0.0119 [−0.010, +0.034]
Mamba→MLP over time	−0.0022 [−0.007, +0.003]	+0.0089 [−0.002, +0.020]
freq Mamba encoder	−0.0024 [−0.011, +0.006]	−0.0117 [−0.044, +0.020]

in *both* directions — is the argument for per-dataset switches over universal defaults, and it does not depend on any single dataset.

Branch matrix: where each domain matters. Because a single Solar cell cannot carry a dual-domain conclusion, we ran the full branch-ablation matrix — full / time-only / freq-only under paired seeds on all nine datasets at H=96 and across all four horizons on the six ETT/Weather/Solar datasets, 27 cells in total (Table 6). Every full-model mean reproduces the corresponding main-table entry, and the Solar H=96 cell replicates the earlier ablation (+0.0126 vs. +0.0114, overlapping CIs). Removing the frequency branch is significant in 5 of 27 cells, concentrated at the shortest horizon: Solar (+0.0126*), ETTm1 (+0.0073*), Electricity (+0.0019*), and Exchange (+0.0015*) at H=96, plus ETTm2 at H=336; Solar's effect does *not* persist at longer horizons. Removing the time branch is significant in 9 of 27 cells, including ETTm1 at *every* horizon (+0.017–+0.021, the most persistent branch effect we measure); on ETTm1, Electricity, and Exchange both branches are individually necessary at H=96. On ETTh1, ETTh2, and Weather no branch-removal effect is detected at *any* of the four horizons, nor on Traffic at H=96. The dual-domain claim is therefore supported but doubly conditional — on the dataset *and* the horizon: the frequency path earns its place mainly at short horizons on four benchmarks and is a no-cost redundancy elsewhere.

Negative results that shaped the design. Deepening the traffic mixer from two to four layers (19.7 → 37.9M) gave no gain, so the release keeps two; enabling the mixer on ETT (8.4K training windows) tripled parameters and was associated with premature early stopping and worse held-out error, so the released ETT configurations disable it.

Table 6. Branch-ablation matrix (27 cells): paired per-seed ΔMSE versus the full model when one branch is removed (three seeds; * t-based 95% CI excludes zero; per-cell CIs ship with the results). Electricity, Traffic, and Exchange were run at H=96 only (—). Full-model means match Table 4.

	freq removed: ΔMSE at H				time removed: ΔMSE at H			
Dataset	96	192	336	720	96	192	336	720
ETTh1	+.0034	+.0005	−.0047	+.0096	−.0007	−.0001	−.0003	+.0176
ETTh2	+.0006	+.0078	+.0054	+.0093	−.0031	−.0017	−.0017	+.0026
ETTm1	+.0073*	+.0002	+.0019	−.0005	+.0196*	+.0168*	+.0203*	+.0210*
ETTm2	+.0020	+.0014	+.0030*	+.0001	+.0060*	+.0060	+.0059*	+.0059
Weather	+.0008	+.0004	−.0002	−.0009	+.0006	+.0011	+.0021	+.0027
Solar	+.0126*	−.0038	+.0006	−.0032	+.0052	+.0075	+.0040*	+.0027
Electricity	+.0019*	—	—	—	+.0017*	—	—	—
Traffic	−.0015	—	—	—	+.0013	—	—	—
Exchange	+.0015*	—	—	—	+.0058*	—	—	—

Table 7. Component ablation on Weather (H=96; MSE mean ± std over three seeds). Δ is the *paired* per-seed difference to the full model with a t-based 95% confidence interval (n=3); * CI excludes zero.

Variant	MSE	paired Δ [95% CI]	Component isolated
full	0.1620±.0003	—	reference
w/o instance norm	0.1781±.0035	+0.0161 [+0.0053, +0.0269]*	RevIN
w/o channel mixer	0.1724±.0021	+0.0104 [+0.0037, +0.0172]*	cross-variate Mamba mixing
Mamba→MLP over time	0.1647±.0005	+0.0027 [+0.0020, +0.0034]*	selective scan (time)
w/o linear backbone	0.1631±.0006	+0.0011 [−0.0014, +0.0036]	DLinear anchor (time)
time branch only	0.1628±.0011	+0.0007 [−0.0020, +0.0035]	frequency branch removed
w/o spectral map	0.1627±.0016	+0.0007 [−0.0041, +0.0055]	FITS-style pathway
freq branch only	0.1626±.0014	+0.0006 [−0.0040, +0.0052]	time branch removed
freq Mamba encoder	0.1626±.0016	+0.0006 [−0.0044, +0.0055]	spectral encoder type
sum fusion	0.1622±.0007	+0.0001 [−0.0020, +0.0023]	learned gate → average

4.4 Analysis

Where each domain wins. The matrix (Table 6) separates three regimes. On Solar the frequency path dominates at H=96 (+0.0126*; removing the time branch is within noise) but its benefit vanishes at longer horizons — consistent with the fact that Solar's 10-minute daily cycle spans 144>L=96 samples, so the spectral path cannot observe a complete daily period and its measured advantage comes from shorter-scale spectral detail, which matters most for near-term steps. On ETTm1, Electricity, and Exchange the branches are complementary at H=96, and on ETTm1 the time branch stays individually necessary at every horizon while the frequency branch's contribution fades beyond H=96. On ETTh1, ETTh2, and Weather (all four horizons) and Traffic the branches are interchangeable, and the convex gate makes carrying the unused path harmless. Which domain a dataset needs — and for how far ahead — is itself dataset-dependent: the dual-domain analogue of the capacity-placement finding.

Data correlations predict — and mislead about — the mixer. A data-mining reading of the placement finding asks *which measurable property of a dataset*

Table 8. Model-only complexity profile of DD-Mamba per dataset (look-back 96, horizon 96, batch 1). GMACs: multiply–accumulates per forecast (`Linear` + `Conv1d` + the selective-scan recurrence); act. mem.: forward activation footprint (fp32). All three columns are hardware-independent estimates; no wall-clock or matched-baseline runtime is claimed.

Dataset	Variates	Params (M)	GMACs	Act. mem. (MB)
Exchange	8	0.09	0.02	2.3
ETTh1	7	0.25	0.08	4.0
Weather	21	5.77	1.90	46.6
Solar	137	0.96	3.26	150.4
Electricity	321	19.70	22.4	254.3
Traffic	862	19.70	60.2	682.7

decides whether cross-variate capacity pays. Mean absolute cross-channel PCC (Table 1) separates the two regimes almost perfectly: the mixer-off ETT family sits at $\overline{|r|} = 0.31$–0.35 (only 14–24% of channel pairs exceed $|r|>0.6$), while the mixer-on Electricity, Traffic, and Solar span 0.50–0.91 (Solar's near-total 0.91 reflects neighboring plants sharing irradiance), matching where the paired ablations find the mixer load-bearing versus harmful. Exchange is the instructive exception: $\overline{|r|} = 0.48$, yet the mixer is disabled — its currency series are near random walks whose correlation is co-trending rather than exploitable cross-sectional signal, and with only 7.6K steps the mixer's parameters cannot be estimated reliably. Raw correlation is therefore a useful *screen*, not a decision rule: high PCC flags candidates for cross-variate capacity, but non-stationarity and sample size can inflate it, and the seed-paired ablation remains the arbiter.

Complexity accounting. Table 8 reports model-only, hardware-independent estimates rather than comparative runtime evidence. Parameter count is governed by model *width*, not channel count (electricity and traffic share 19.7M despite a 2.7× gap in variates), so model size is decoupled from data dimensionality; compute and activation memory instead scale with the channel count through the variate mixer (traffic 60 GMACs vs. electricity's 22 at equal width, while ETT and exchange sit two orders of magnitude below), keeping the capacity restraint on those datasets inexpensive. Wall-clock throughput and matched-baseline runtime remain unmeasured.

5 Conclusion

DD-Mamba combines a DLinear-style temporal backbone, a selective-state-space correction, an optional frequency pathway, and dataset-dependent cross-variate mixing. Against quoted results under the look-back-96 protocol, it shows average-MSE gaps above the descriptive tolerance on Solar, ETTh1, and ETTh2 (only Solar and ETTh1 also clearing the wider 0.01 cross-paper caution), lies within that tolerance of the strongest reported result on four datasets, and trails on Exchange and Traffic. The ablations support two conclusions. First, *placement*: the variate mixer helps Weather yet hurts ETTh1 and normalization helps Weather yet hurts Solar, each direction backed by a paired CI excluding zero on a different dataset. Second, the 27-cell branch matrix locates the dual-domain benefit in both dataset and horizon: the frequency branch is individually significant on Solar, ETTm1, Electricity, and Exchange at H=96 but fades at longer horizons, the time branch stays necessary on ETTm1 at every horizon, and the branches are redundant on ETTh1, ETTh2, and Weather across all four horizons and on Traffic. Zero initialization is accuracy-neutral and retained only for its controlled linear starting point. These findings establish conditional component placement: dual-domain value is real mainly at short horizons on four of nine benchmarks and structurally harmless elsewhere, not a universal gain.

The evidence remains limited in several ways. Baselines other than DLinear on ETTh1 are quoted rather than rerun in one pipeline; Electricity, Traffic, and Exchange enter the branch matrix at H=96 only, the other component ablations cover only H=96, all at three seeds without family-wise correction; schedule sensitivity and head-norm dynamics are not measured; and the complexity table lacks wall-clock and matched-baseline runtime. The fixed 96-step look-back also leaves the Traffic weekly-cycle hypothesis untested, while Exchange at H=720 remains a notable failure case. Explicit forecasting of time-varying dispersion is left as future work rather than claimed as part of the evaluated model.

Reproducibility. Code, configurations, and the ablation harness are supplementary material, to be released publicly upon acceptance.

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